

Question				Marking details	Marks Available																	
					AO1	AO2	AO3	Total	Maths	Prac												
2.	(a)	(i)		Large surface area (relative to volume of organism)	1			1														
		(ii)		{longer diffusion distance/ use of cross section data} in Lumbricus (1) - so diffusion alone cannot meet (O₂/ metabolic) demands / OWTTE (1) {circulatory system / Hb/ blood} can deliver <u>oxygen</u> {to tissues / cells/ over greater distances}. (1)	3			3														
		(iii)		Tracheal system / tracheoles (1) ignore trachea <u>Oxygen</u> delivered <u>directly</u> to {tissues/ cells/ muscles} (1) Ignore reference to open circulation	2			2														
	(b)	(i)		Increase surface area (: vol ratio)/ contain capillary (network) (1)			1	1														
		(ii)		Any five (×1) from: <table><tr><td><i>Nereis</i></td><td><i>Lumbricus</i></td></tr><tr><td>A. {fast-moving/ predator} so has a greater O₂ requirement /</td><td>{slow-moving/ detritivore} so has a smaller O₂ requirement (1)</td></tr><tr><td>B. <u>Hb</u> lower affinity for O₂ /</td><td><u>Hb</u> higher affinity for O₂ (1)</td></tr><tr><td>C. <u>Hb</u> dissociates {more readily/ more efficiently/ at higher pO₂}/ more oxygen released (1) reject faster</td><td>E. <u>Hb</u> {fully saturated/ higher saturation} at lower pO₂ (1)</td></tr><tr><td>D. to respiring tissues. (1)</td><td>F. As habitat has low oxygen availability (1)</td></tr></table>	<i>Nereis</i>	<i>Lumbricus</i>	A. {fast-moving/ predator} so has a greater O ₂ requirement /	{slow-moving/ detritivore} so has a smaller O ₂ requirement (1)	B. <u>Hb</u> lower affinity for O ₂ /	<u>Hb</u> higher affinity for O ₂ (1)	C. <u>Hb</u> dissociates {more readily/ more efficiently/ at higher pO ₂ }/ more oxygen released (1) reject faster	E. <u>Hb</u> {fully saturated/ higher saturation} at lower pO ₂ (1)	D. to respiring tissues. (1)	F. As habitat has low oxygen availability (1)				3	2	5		
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				Question 2 total	6	3	3	12	0	0												